

# CT214H – Web Programming

## Practical Session 5 – MySQL & PHP

### Content

- ❖ PHP - MySQL: continue add features
- ❖ Prerequisites: XAMPP installed. Your starting point should be from the Techblog web page in the previous practical session.
- ❖ **Objective:**
  - Connect to database
  - Query data and display on your page
- ❖ **Requirement:**
  - Step-by-step do the tasks below and observe the results

### 1. Display popular articles

In your techblog.php, display most viewed articles in sidebar

- In the popular articles section (inside your side bar): query the 5 most popular articles ordered by views

```
$popular_sql = "SELECT id, title, views FROM articles ORDER BY views DESC LIMIT 5";  
$popular_result = $conn->query($popular_sql);
```

- Change the html to display the query results

```
<div class="widget popular">  
  <h3>Popular Articles</h3>  
  <ul class="popular-list">  
    <?php while($pop = $popular_result->fetch_assoc()): ?>  
      <li>  
        <a href="article.php?id=<?php echo $pop['id']; ?>">  
          <?php echo $pop['title']; ?> </a>  
      </li>  
    <?php endwhile; ?>  
  </ul>  
</div>
```

### 2. Pagination

a. Adding more sample data into your articles table: [sample1.txt](#)

b. In your techblog.php, display 6 articles per page with next/previous buttons

- Inside of your article grid section, changing the SQL to query only articles for a page

```
// Include database connection  
include 'db-connect.php';  
// Pagination settings  
$articles_per_page = 6;
```

```
// Get current page number from URL, default to 1
$current_page = isset($_GET['page']) ? (int)$_GET['page'] : 1;
$current_page = max(1, $current_page); //Ensure page is at least 1
// Calculate offset for SQL query
$offset = ($current_page - 1) * $articles_per_page;
// Count total articles
$count_sql = "SELECT COUNT(*) as total FROM articles";
$count_result = mysqli_query($conn, $count_sql);
$count_row = mysqli_fetch_assoc($count_result);
$total_articles = $count_row['total'];
// Calculate total pages
$total_pages = ceil($total_articles / $articles_per_page);
// Query to get all articles
$sql = "SELECT * FROM articles LIMIT $articles_per_page OFFSET $offset";
$result = mysqli_query($conn, $sql);
```

- Keep the loop to display the article grids unchanged.
- Below the article grid section, add another section to control pagination

```
<!-- Pagination Controls -->
<div class="pagination">
  <!-- Previous Button -->
  <?php
  if ($total_pages > 1){
    if ($current_page > 1){ ?>
      <a href="?page=?php echo $current_page - 1; ?>" class="pagination-btn prev-btn">< Previous</a>
    <?php }else{ ?>
      <span class="pagination-btn prev-btn disabled">< Previous</span>
    <?php } ?>
  <!-- Page Numbers -->
  <div class="page-numbers">
    <?php
    // Show page numbers
    for ($i = 1; $i <= $total_pages; $i++) {
      if ($i == $current_page) {
        echo "<span class='page-number active'>$i</span>";
      } else {
        echo "<a href='?page=$i' class='page-number'>$i</a>";
      }
    }
  }?>
</div>
```

```

<!-- Next Button -->
<?php
if ($current_page < $total_pages){ ?>
    <a href="?page=<?php echo $current_page + 1; ?>" class="pagination-btn next-btn">Next →</a>
<?php }else{ ?>
    <span class="pagination-btn next-btn disabled">
        Next →</span>
<?php
    }
}??>
</div>

```

c. Styling the pagination section: [pagination.css](#)

d. Changing your category.php to enable pagination (Optional)

- Inside of your category.php page, add the pagination setting as above to calculate variables such as: current\_page, total\_pages offset, etc.

- Change the SQL to query only articles for the current page

```

$sql = "SELECT * FROM articles WHERE category = '{$category['name']}' LIMIT $articles_per_page
OFFSET $offset";

```

- Below the article grid section, add the pagination controls as previous example

### 3. Related articles

Show 3 related articles (same category) in your article.php

- Before the closing main tag in your page, add a section for related articles

```

<section class="related-articles-section">
<h3><img alt="Related Articles icon" data-bbox="171 558 191 571"/> Related Articles</h3>
<div class="related-articles-grid">
<?php
// Get 3 related articles from same category, excluding current article
$related_sql = "SELECT id, title, image_url, excerpt, author, created_at, views FROM articles WHERE
category = '{$article['category']}'
AND id != $article_id ORDER BY created_at DESC LIMIT 3";
$related_result = mysqli_query($conn, $related_sql);
if (mysqli_num_rows($related_result) > 0) {
    while($related = mysqli_fetch_assoc($related_result)) { ?>
        <article class="related-card">
            "
                class="related-image">
            <div class="related-content">
                <h4 class="related-title">
                    <?php echo htmlspecialchars($related['title']); ?>
                </h4>

```

```

        <p class="related-excerpt">
            <?php echo substr(htmlspecialchars($related['excerpt']), 0, 100) . '...'; ?>
        </p>
        <div class="related-meta">
            <span>By <?php echo htmlspecialchars($related['author']); ?></span>
            <span class="date"><?php echo $related['created_at']; ?> </span>
        </div>
        <a href="article.php?id=<?php echo $related['id']; ?>" class="read-more">Read More →</a>
        <p class="view-count" style="color: #999; font-size: 0.9rem; margin-top: 10px;"> 👁
    <span class="views"> <?php echo $related['views']; ?> </span> views
    </p>
</div>
</article>
<?php
}
} else {
    echo '<p class="no-related">No related articles found.</p>';
}
?>
</div>
</section>

```

- Update your article.css file, add these styles at the end of the file: [article-related.css](#)

#### 4. Add new article form

Create add article form page

a. Make sure you have a folder name 'images' in your project directory

On Linux/Mac:

- mkdir images
- chmod 755 images

On Windows: Just create the folder normally.

Inside of the navigation bar in techblog.php, add an item to add new article

```
<li><a href="add-article.php">Add New Article</a></li>
```

b. Create add-article.php

- Copy the techblog.php, name as add-article.php

- Change the main container (inside main tag) to contain uploading form

```
<main class="upload-form-container"> </main>
```

- Inside the main container create a div upload card

```
<div class="upload-form-card"></div>
```

- Inside the div, add the following content:

+ An h1 for adding new article

```
<h1 class="upload-form-title"><img alt="pencil icon" data-bbox="358 181 375 197"/> Add New Article</h1>
```

+ Announcement for success message

```
<?php if ($success): ?>
<div class="upload-message upload-message-success">
  ✓ Article published successfully!
  <a href="article.php?id=<?php echo $new_article_id; ?>">View Article</a>
  or
  <a href="techblog.php">Go to Home</a>
</div>
<?php endif; ?>
```

+ Announcement for error message

```
<?php if ($error): ?>
<div class="upload-message upload-message-error">
  ✗ <?php echo htmlspecialchars($error); ?>
</div>
<?php endif; ?>
```

+ An form for article submission

```
<form method="POST" action="upload-article.php" enctype="multipart/form-data" id="articleForm">
</form>
```

+ Inside the form submission, add the following content

→ A section for upload image

```
<div class="upload-form-group">
<label>Article Image <span class="required">*</span></label>
<div class="image-upload-area" id="uploadArea">
  <div class="upload-icon"><img alt="upload icon" data-bbox="335 668 352 685"/></div>
  <p class="upload-text">Click to upload or drag and drop</p>
  <p class="upload-hint">JPG, PNG, GIF, WEBP (Max 5MB)</p>
  <input type="file"
    name="article_image"
    id="articleImage"
    class="file-input-hidden"
    accept="image/jpeg,image/png,image/gif,image/webp"
    required>
</div>
```

```

<!-- Image Preview -->
<div class="image-preview-container" id="imagePreviewContainer">
  <img src="" alt="Preview" class="image-preview" id="imagePreview">
  <div class="image-info" id="imageInfo"></div>
  <button type="button" class="remove-image-btn" id="removeImageBtn">Remove Image</button>
</div>
</div>

```

→ A section for adding title

```

<div class="upload-form-group">
  <label for="title">
    Article Title <span class="required">*</span>
  </label>
  <input type="text" id="title" name="title" class="upload-form-input" placeholder="e.g., Getting
  Started with Machine Learning"
  required value="<?php echo isset($_POST['title']) ? htmlspecialchars($_POST['title']) : ''; ?>"
  <span class="upload-input-hint">Choose a clear, descriptive title</span>
</div>

```

→ A section for selecting category loaded from database

```

<div class="upload-form-group">
  <label for="category">
    Category <span class="required">*</span>
  </label>
  <select id="category" name="category" class="upload-form-select" required>
    <option value="">-- Select Category --</option>
    <?php
    // Fetch categories from database
    $cat_sql = "SELECT id, name FROM categories ORDER BY name ASC";
    $cat_result = mysqli_query($conn, $cat_sql);
    // Check if categories exist
    if (mysqli_num_rows($cat_result) > 0) {
      while ($cat = mysqli_fetch_assoc($cat_result)) {
        $selected = (isset($_POST['category']) && $_POST['category'] == $cat['name']) ? 'selected' : '';
        echo '<option value="' . htmlspecialchars($cat['name']) . "' . $selected . '>'
        . htmlspecialchars($cat['name']) . '</option>';
      }
    }
  </select>

```

```

    } else {
        echo '<option value="" disabled>No categories available</option>';
    }
    ?>
</select>
<span class="upload-input-hint">Select the category that best fits your article</span>
</div>

```

→ A section for adding author

```

<div class="upload-form-group">
    <label for="author">
        Author Name <span class="required">*</span>
    </label>
    <input type="text" id="author" name="author" class="upload-form-input" placeholder="e.g., John Doe" required
        value="<?php echo isset($_POST['author']) ? htmlspecialchars($_POST['author']) : ''; ?>">
</div>

```

→ A section for adding published date

→ A section for adding excerpt

```

<div class="upload-form-group">
    <label for="created_at">
        Published Date <span class="required">*</span>
    </label>
    <input type="date" id="created_at" name="created_at"
        class="upload-form-input" required
        value="<?php echo isset($_POST['created_at']) ? $_POST['created_at'] : date('Y-m-d'); ?>">
</div>

```

```

<div class="upload-form-group">
    <label for="excerpt">
        Article Excerpt <span class="required">*</span>
    </label>
    <textarea id="excerpt" name="excerpt" class="upload-form-textarea"
        placeholder="Write a brief summary (2-3 sentences)"
        required><?php echo isset($_POST['excerpt']) ? htmlspecialchars($_POST['excerpt']) : '';
    ?></textarea>
    <span class="upload-input-hint">This will appear on the article cards</span>
</div>

```

→ A section for adding article content

```
<div class="upload-form-group">
  <label for="content">
    Full Article Content <span class="required">*</span>
  </label>
  <textarea id="content" name="content" class="upload-form-textarea"
    style="min-height: 300px;"
    placeholder="Write your full article content here..."
    required><?php echo isset($_POST['content']) ? htmlspecialchars($_POST['content']) : '';
  ?></textarea>
  <span class="upload-input-hint">Write your complete article here</span>
</div>
```

→ A section for submission button

```
<div class="upload-form-buttons">
  <button type="submit" class="upload-btn-submit" id="submitBtn">
     Publish Article
  </button>
  <a href="index.php" class="upload-btn-cancel">
    Cancel
  </a>
</div>
```

- In the beginning of your page, connect to database to store new article

```
require_once 'db-connect.php';
// Initialize variables
$success = false;
$error = "";
$new_article_id = 0;
// Process form submission
if ($_SERVER['REQUEST_METHOD'] == 'POST') {...}
```

- Inside of the form submission processing, get form data, validate, and insert into the articles table

+ Get form data

```
// Get and sanitize form data
$title = mysqli_real_escape_string($conn, trim($_POST['title']));
$excerpt = mysqli_real_escape_string($conn, trim($_POST['excerpt']));
$content = mysqli_real_escape_string($conn, trim($_POST['content']));
$category = mysqli_real_escape_string($conn, trim($_POST['category']));
$author = mysqli_real_escape_string($conn, trim($_POST['author']));
$created_at = mysqli_real_escape_string($conn, $_POST['created_at']);
```



#### + Handle image

```
$image_url = '';
$upload_success = false;
if (isset($_FILES['article_image']) && $_FILES['article_image']['error'] == 0) {
    $file = $_FILES['article_image'];
    // Get file info
    $file_name = $file['name'];
    $file_tmp = $file['tmp_name'];
    $file_size = $file['size'];
    $file_error = $file['error'];
    // Get file extension
    $file_ext = strtolower(pathinfo($file_name, PATHINFO_EXTENSION));
    // Allowed extensions
    $allowed_extensions = array('jpg', 'jpeg', 'png', 'gif', 'webp');
    // Check if extension is allowed
    if (in_array($file_ext, $allowed_extensions)) {
        // Check file size (max 5MB)
        if ($file_size <= 5242880) {
            // Generate unique filename
            $new_filename = 'article_' . uniqid() . '.' . $file_ext;
            // Upload directory
            $upload_dir = 'images/';
            $upload_path = $upload_dir . $new_filename;
            // Move uploaded file
            if (move_uploaded_file($file_tmp, $upload_path)) {
                $image_url = $upload_path;
                $upload_success = true;
            } else {
                $error = "Failed to upload image. Please check folder permissions.";
            }
        } else {
            $error = "File size too large. Maximum size is 5MB.";
        }
    } else {
        $error = "Invalid type. Allowed: JPG, JPEG, PNG, GIF, WEBP.";
    }
} else {
    $error = "Please select an image to upload.";
}
```

#### + Validate required fields

```
if (empty($title) || empty($excerpt) || empty($content) || empty($category) || empty($author) ||
empty($created_at)) {
    $error = "Please fill in all required fields.";
} elseif ($upload_success) {
```

#### + Insert into articles table

```
// Insert into database
$sql = "INSERT INTO articles (title, excerpt, content, category, author, created_at, image_url)
VALUES ('$title', '$excerpt', '$content', '$category', '$author', '$created_at', '$image_url')";
if (mysqli_query($conn, $sql)) {
    $success = true;
    $new_article_id = mysqli_insert_id($conn);
} else {
    $error = "Database Error: " . mysqli_error($conn);
    // Delete uploaded image if database insert fails
    if (file_exists($image_url)) {
        unlink($image_url);
    }
}
}
```

- c. Add a JavaScript file to handle image upload in your php: [add-article.js](#)
- d. Link to style sheet to style your page: [add-article.css](#)

## 5. Update an article

- a. Inside of the article.php adding a link to edit the article

Add this near the end of article content, before closing `</div>`, using the same style as the Back to articles button has.

```
<a href="edit-upload-article.php?id=<?php echo $article_id; ?>" class="back-to-articles">Edit Article</a>
```

- b. Create edit-upload-article.php

Same as previous exercise, you only need to change few things to edit an article:

- Copy the techblog.php, name as edit-upload-article.php
- Change the main container (inside main tag) to contain edit form: [main-edit-form](#)
- In the beginning of your page, connect to database to update article: [edit-article.php](#)

- c. Add a JavaScript file to handle image upload in your php: [edit-upload-article.js](#)
- d. Link to style sheet to style your page: [edit-upload-article.css](#)

## 6. Delete an article

- a. Inside of the article.php adding a link to delete the article

Add this near the end of article content, before closing `</div>`, using the same style as the Back to articles button has.

```
<a href="delete-article.php?id=?php echo $article_id; ?>" class="back-to-articles"
  onclick="return confirm('Are you sure you want to delete this article? This action cannot be
  undone!');">
   Delete Article
</a>
```

b. Create a [delete-article.php](#) to handle the database deletion

This need to delete the article in database and also delete the image in your file system.

c. Showing success message in your techblog.php

Add this right after <body> or before articles

```
if (isset($_GET['deleted']) && $_GET['deleted'] == 'success') {
  echo "<script>alert('Article deleted successfully!');</script>";
}
```

At this moment, you should have a basic development of your website!